Supplement S3: Observable Dataset

33 Relations Compared with Measurement; 62 Conditional and Diagnostic Entries

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S3.1 Introduction

This supplement documents the framework's observable dataset (`canonical/data/observables.json`, 95 entries) and the comparison data behind the main text. The dataset is a ledger, not a list of predictions. It counts **33 relations** (Type I) as the framework's claims, compared with measurement in main §4. The other **62 entries** (Types II–IV) are conditional extractions, outputs of mechanisms that consume calibration choices, or diagnostics of the approximate seam metric; they are kept here for traceability and are not counted (main §4.5).

The 33 relations are closed forms in twenty declared constants, assigned to observables by the author (main §3); the metric enters none of them. Each entry below carries a status that says what it is. The labels R1 to R12 used in the statuses are the items of the residue register of main §3.3, the list of what the framework sets rather than derives.

S3.2 Type classification

Type	Count	What it is	Counted in main §4
I	33	Dimensionless closed forms in the structural constants, assigned to measured parameters	yes

Type	Count	What it is	Counted in main §4
II	19	Dimensionful numbers read in a chosen unit (register item R10, the unit of dimensionful entries), scale bridges, one ratio of masses taken at different scales and its product with m_s/m_d, and legacy values without a documented derivation	no
III	21	Outputs of mechanisms on the approximate seam metric (Wilson-line splitting, associative volumes, spectral counts, bundle conditioning), with calibration choices; and four retired entries of a supersymmetric running	no
IV	22	Structural quantities and numerical diagnostics, with no experimental counterpart	no

Type I. Each relation is a ratio of structural integers or uses one further operation ($\sqrt{}$, φ, ln 2, ζ values), declared in its status. The status vocabulary is that of main §4.1: *integer*, *algebraic / φ / ln 2 / ζ values*, *det(g)* (the imposed normalization, R2), κ_T (the integer combination 61, R3), *proximity*, *tension*, *approximate* (one of several incompatible relations in an overdetermined sector: charged leptons, R8; cosmological densities, R11; CKM, R12), *inside the PDG range*, *unit assignment* (the unit of δ_{CP}, R6); m_b/m_t is compared in the mass scheme declared in register item R7 (comparison scale and mass scheme).

Type II. Several entries are numbers read in MeV or GeV, for example $m_u = \sqrt{}(\dim G_2/3) = 2.160$, $m_d = \ln 107 = 4.673$, $m_c = (\dim G_2 - \pi)^3 = 1280.3$, $m_t = 415^2$. Reading a structural number in a unit is an assignment of the same kind as reading 197 in degrees, with a far larger space of alternatives (R10). The six CKM magnitudes and the ratios m_b/m_d and m_t/m_c are legacy values: their derivation is not documented in the current dataset, and they are retained for traceability. These two ratios are not fixed by the quark ratios of the 33 relations alone, which relate two separate groups, {u, d, s, c} and {b, t}. m_c/m_d is the product of the mixed-scale m_c/m_s, of Type II, with m_s/m_d.

Type III. The Wilson-line and associative-volume entries reproduce lepton mass ratios at the percent level with declared calibration choices (a coupling c = 0.452 and a cycle assignment) selected against the measured ratios, and an exponent $\alpha = e^K = 0.002763$ computed from the Kähler potential of the approximate metric; they are conditional reconstructions (main §6.2). The comparison of the volume difference $\Delta V(e\text{-}\mu)$ is mismatched: its value is $Vol_e - Vol_\mu = 3.271$, its reference is $\ln(m_\tau/m_\mu) = 2.823$, and the instanton relation pairs that reference with $Vol_\mu - Vol_\tau = 5.362$ (the matching reference for $Vol_e - Vol_\mu$ is $\ln(m_\mu/m_e) = 5.33$), so its 15.9% deviation carries no meaning. The cycle search scored this same pair. The combined m_τ/m_μ uses the same difference, and the combined m_μ/m_e is the ratio of the two combined entries; with $Vol_\mu - Vol_\tau$ they would read 16.79 and 207.6, still within 1%. The entries are kept as computed. The four entries of a two-loop supersymmetric running from a grand-unification boundary are retired: the framework uses no running between scales (main §5.3). The remaining entries are diagnostics, some compared with a mathematical expectation (the Weyl exponent with d/2, the gauge-kinetic conditioning with 1), none with a measurement.

Type IV. Topological inputs (the posited Betti pair b_2, b_3: R1), integer combinations (H^*, N_{gen}), the imposed normalization det(g) = 65/32 (R2), numerical diagnostics of the certified seam metric (Newton–Kantorovich quantities, Gram conditioning, spectral counts, metric block eigenvalues), and three entries taken from the black-hole-remnant analysis of main ref. [42], one of which depends on the scale identification $\tau_0 = v_{EW}$.

S3.3 Experimental data and conventions

The comparison data are those of the most recent releases (script `canonical/scripts/k7_v3_6_exp_refresh.py`, log `canonical/results/k7_v3_6_exp_refresh.json`):

Source	Used for	Convention
PDG 2026 [1]	particle masses, $\sin^2\theta_W$ and α_s at M_Z, CKM parameters, quark mass ratios	MS-bar at the PDG reference scales; m_t from direct measurements
NuFIT 6.1 [27]	neutrino mixing	normal ordering, with the tabulated Super-Kamiokande and IceCube atmospheric data; one variant for all four entries
Planck 2018 [3]	cosmology	base ΛCDM, TT,TE,EE+lowE+lensing; one column for all entries; the PR4 reanalysis [49] is not used
CODATA 2022 [50]	α at $q^2 = 0$, m_μ/m_e	n/a

Derived reference values. Mass ratios are computed from the published masses, with error propagation that neglects correlations. This is an approximation, not a bound: the summary tables do not publish

the covariances, and for a ratio R = x/y the term -2 Cov(x, y)/(xy) in Var(R)/R^2 can have either sign. The rule holds for every mass ratio, m_W/m_Z included: it is computed from m_W = 80.3625(77) GeV and m_Z = 91.1879(20) GeV, which gives 0.88128(9); the ratio line of the PDG summary table, 0.88136(15), is not used. The six density ratios (Ω_{DM}/Ω_b, Ω_c/Ω_Λ, Ω_Λ/Ω_m, Ω_b/Ω_m, Ω_c/Ω_m, Ω_m/Ω_Λ) are computed sample by sample on the published Planck chains [3] (base_plikHM_TTTEEE_lowl_lowE_lensing, 25 225 weighted samples, whose marginal statistics reproduce the published table), as weighted mean and standard deviation; `canonical/scripts/k7_v3_6_planck_ratios.py`. No first-order propagation or proposal covariance enters. The correlations matter: ω_c and ω_m are correlated at about 0.99, and Ω_c/Ω_m is 0.8390(19) on the chains, at $+2.5\sigma$, against 0.839(11) when they are neglected.

Total matter. Planck's Ω_m includes neutrinos (Ω_ν h^2 = 0.00064 at Σm_ν = 0.06 eV, Ω_ν/Ω_m = 0.0045). So Ω_b/Ω_m + Ω_c/Ω_m = 0.9955 in the reference, while the formulas 5/32 and 27/32 sum to 1. This is a defect of compatibility with the Planck definition; the denominator is not replaced by Ω_b + Ω_c, since that would change the quantities tested. For information, that other quantity gives 0.8428(19) for $\Omega_c/(\Omega_b$ + $\Omega_c)$, and pulls of $\pm 0.5\sigma$. The complementarity of baryons and cold dark matter, with neutrinos neglected, does not require flatness. λ_H is the tree-level $m_H{}^2/(2v^2)$ with $v = (\sqrt{2}\ G_F)^{-1/2}$, a definition that is itself a scheme choice. m_s/m_d is given by the PDG as the range 17–22, without an uncertainty; it is stored as a range, and no pull is defined (the relation's value 20 lies inside it).

Confidence levels. All uncertainties are brought to 1σ. The PDG summary tables quote the light-quark masses, m_c, m_b and m_u/m_d at 90% confidence; these are divided by 1.645. The lattice average of m_c/m_s (PDG quark-mass review, statistical and systematic errors in quadrature) is already a 1σ value.

Asymmetric uncertainties. NuFIT and the CKM parameter A have asymmetric errors; both sides are kept in the dataset, and the pull uses the side facing the framework's value. The $\sin^2\theta_{23}$ profile is not Gaussian: its 3σ range (0.435–0.584) contains 6/11 although the Gaussian pull exceeds 4σ (main §4.3).

Classes. With one convention for all uncertainties, the 33 relations fall into the five classes of main §4.2: 23 within 2σ, 2 in tension, 6 proximities beyond 3σ (among them m_b/m_t, in the mass scheme of R7, and m_W/m_Z), 1 inside the PDG range, 1 with a non-Gaussian profile.

Not refreshed. Y_p (0.245 ± 0.003) is the PDG's recommended observational helium abundance from HII regions (BBN review of the 2024 Review of Particle Physics [56], eq. 24.3); it was not re-checked against the 2026 edition. The Type II CKM magnitudes and mass products, and three Type III entries (the retired α_s of the running and the two associative-volume differences), keep earlier reference values (PDG 2025), since they are not compared in the main text.

S3.4 Structural constants

The twenty constants, their origin and their status are given in main §3.1. N_{gen} = 3 is an integer identity (b_2/dim K_7; main §2.4). b_2 and b_3 are inputs posited by the framework (R1); the spectral analysis of [B] concerns an approximate metric and does not confirm them.

S3.5 The 33 relations: reference data

Formulas, framework values and deviations are in main §4.1. This table gives, for each relation, the reference value, its source and the convention under which it is compared. It is generated from the dataset (`canonical/scripts/k7_v3_6_s3_tables.py`); the status column is read from main §4.1.

ID	Observable	Measured	Source	Pull	Convention / note	Status
A01	$\sin^2\theta_W$ (M_Z)	0.23122(6)	PDG 2026	-7.5σ	MS-bar at M_Z (PDG Table 1.1)	integer; **proximity**
A02	$\alpha_s(M_Z)$	0.1180(9)	PDG 2026	-0.2σ	n/a	algebraic
A03	λ_H	0.12914(23)	PDG 2026	-1.3σ	tree level ${m_H}^2/(2v^2)$, v = $(\sqrt{2}\,G_F)^{-1/2}$ = 246.220 GeV	algebraic; det(g)
A04	$\alpha^{-1}(0)$	137.035999177(21)	CODATA 2022	$\gg 10\sigma$	α^{-1} at $q^2 = 0$	det(g), κ_T; **proximity**
A05	Q_{Koide}	0.6666645(51)	PDG 2026	$+0.4\sigma$	computed from the PDG 2026 masses	integer; approximate (R8)
A06	m_τ/m_e	3477.37(18)	PDG 2026	-2.1σ	from the PDG 2026 masses	integer; approximate (R8); **tension**
A07	m_μ/m_e	206.7682827(46)	CODATA 2022	$\gg 10\sigma$	n/a	φ; approximate (R8); **proximity**
A08	m_s/m_d	17–22 (range)	PDG 2026	n/a	PDG gives the range 17–22, not a measurement with an uncertainty	integer; inside the PDG range
A09	m_c/m_s	11.769(35)	PDG 2026	-1.6σ	lattice average (N_f = 2+1+1), PDG quark-mass review	integer
A10	m_b/m_t	0.024253(43)	PDG 2026	-10.2σ	m_b at 90% CL converted to 1σ	integer; **proximity** (in the scheme declared in R7)
A11	m_u/m_d	0.462(12)	PDG 2026	$+0.7\sigma$	MS-bar at 2 GeV	integer
A12	$\sin^2\theta_{12}$ (PMNS)	0.3088(+67/−66)	NuFIT 6.1 (IC24+SK-atm, NO)	-0.2σ	n/a	integer
A13	$\sin^2\theta_{23}$ (PMNS)	0.470(+17/−14)	NuFIT 6.1 (IC24+SK-atm, NO)	$+4.4\sigma$*	3σ range 41.27°–49.86°, i.e. $\sin^2$ 0.435–0.584, contains 6/11	integer; **tension** (octant; inside the NuFIT 3σ range, main §4.3)
A14	$\sin^2\theta_{13}$ (PMNS)	0.02248(+55/−59)	NuFIT 6.1 (IC24+SK-atm, NO)	-0.5σ	n/a	integer
A15	δ_{CP}	212 (+26/−36)°	NuFIT 6.1 (IC24+SK-atm, NO)	-0.4σ	degrees	**unit assignment** (R6)
A16	$\sin\theta_{12}$ (CKM)	0.22517(68)	PDG 2026	$+0.9\sigma$	PDG CKM review: $\sin\theta_{12} = \lambda$ (not squared)	integer; approximate (R12)
A17	A (Wolfenstein)	0.826(+17/−15)	PDG 2026	$+0.7\sigma$	PDG CKM review, Wolfenstein fit	integer; approximate (R12)
A18	$\sin\theta_{23}$ (CKM)	0.04189(+81/−69)	PDG 2026	-0.3σ	PDG CKM review: $\sin\theta_{23} = A\lambda^2$ (not squared)	integer; approximate (R12)
A19	m_H/m_t	0.7250(13)	PDG 2026	$+1.8\sigma$	m_t from direct measurements	integer
A20	m_H/m_W	1.5571(14)	PDG 2026	$+0.5\sigma$	n/a	integer
A21	m_W/m_Z	0.881285(87)	PDG 2026	-3.8σ	from the PDG 2026 masses	integer; **proximity**
A22	m_μ/m_τ	0.0594612(30)	PDG 2026	$+20.8\sigma$	from the PDG 2026 masses	integer; approximate (R8); **proximity**

ID	Observable	Measured	Source	Pull	Convention / note	Status
A23	Ω_{DE}	0.6847(73)	Planck 2018 (TT,TE,EE +lowE + lensing)	$+0.2\sigma$	Ω_Λ	ln 2; approximate (R11)
A24	n_s	0.9649(42)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	n/a	ζ values
A25	Ω_{DM}/Ω_b	5.364(77)	Planck 2018 (TT,TE,EE +lowE + lensing)	$+0.1\sigma$	$\Omega_c\ h^2$ / $\Omega_b\ h^2$	integer; approximate (R11)
A26	Ω_c/Ω_Λ	0.387(14)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	$\Omega_c\ h^2$ / h^2 / Ω_Λ	integer; det(g); approximate (R11)
A27	Ω_Λ/Ω_m	2.173(74)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	Ω_Λ / Ω_m, Planck definitions	integer; approximate (R11)
A28	h	0.6736(54)	Planck 2018 (TT,TE,EE +lowE + lensing)	0.0σ	n/a	integer
A29	Ω_b/Ω_m	0.1565(19)	Planck 2018 (TT,TE,EE +lowE + lensing)	-0.1σ	$\Omega_b\ h^2$ / $\Omega_m\ h^2$, Ω_m includes neutrinos	integer; det(g); approximate (R11)
A30	Ω_c/Ω_m	0.8390(19)	Planck 2018 (TT,TE,EE +lowE + lensing)	$+2.5\sigma$	$\Omega_c\ h^2$ / $\Omega_m\ h^2$, Ω_m includes neutrinos	integer; det(g); approximate (R11); **tension**
A31	σ_8	0.8111(60)	Planck 2018 (TT,TE,EE +lowE + lensing)	-0.3σ	n/a	integer; det(g)
A32	Y_p	0.245(3)	PDG 2024 (BBN review, eq. 24.3)	$+0.3\sigma$	n/a	integer; κ_T

ID	Observable	Measured	Source	Pull	Convention / note	Status
A33	Ω_m/Ω_Λ	0.461(16)	Planck 2018 (TT,TE,EE +lowE + lensing)	-0.1σ	Ω_m / Ω_Λ, Planck definitions	integer; κ_T; approximate (R11)

* Non-Gaussian profile (S3.3).

S3.6 The 62 conditional and diagnostic entries

Generated from the dataset. "Reference value" is the measured value where one exists, otherwise the mathematical expectation the entry is compared with, otherwise n/a. None of these entries is counted as a prediction.

ID	Type	Quantity	Formula or mechanism	Framework value	Reference value	Status
B01	II	m_u	sqrt(dim(G_2)/3)	2.16025	2.16	unit assignment (R10)
B02	II	m_d	ln(107)	4.673	4.7	unit assignment (R10)
B03	II	m_s	$\tau\times24$	93.52	92.9	unit assignment (R10)
B04	II	m_c	(dim(G_2)-π)^3	1280.26	1272.9	unit assignment (R10)
B05	II	m_b	$2b_2\times H^*$	4158	4186	unit assignment (R10)
B06	II	m_t	415^2	172225	172600	unit assignment (R10)
B07	II	v_{EW}	Scale bridge from the unit assignment (R10)	246.87	246.22	scale bridge; unit assignment (R10)
B08	II	M_W	$(g_2/2)\times v_{EW}$	80.4	80.3625	scale bridge; unit assignment (R10)
B09	II	M_Z	$M_W/\cos(\theta_W)$	91.2	91.1879	scale bridge; unit assignment (R10)
B10	II	$\lvert V_{us}\rvert$	not documented (legacy label: Cabibbo from b_2)	0.2245	0.2243	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B11	II	$\lvert V_{cb}\rvert$	not documented (legacy label: 2nd gen mixing)	0.04214	0.0422	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B12	II	$\lvert V_{ub}\rvert$	not documented (legacy label: 3rd gen coupling)	0.003947	0.00394	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B13	II	$\lvert V_{td}\rvert$	not documented (legacy label: down-type mixing)	0.008657	0.00867	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B14	II	$\lvert V_{ts}\rvert$	not documented (legacy label: strange-top coupling)	0.04154	0.0415	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B15	II	$\lvert V_{tb}\rvert$	not documented (legacy label: unitarity)	0.999106	0.999105	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction

ID	Type	Quantity	Formula or mechanism	Framework value	Reference value	Status
B16	II	m_c/m_s (ext)	$\tau\times3.49$	13.6	13.6	mixed scales (m_c at m_c, m_s at 2 GeV); not comparable
B17	II	m_c/m_d	m_c/m_s (ext, B16) × m_s/m_d (A08) = $\tau\times3.49\times20$	271.99	272	cascade of B16 (Type II, not comparable) and A08; no independent content
B18	II	m_b/m_d	not documented (legacy label: cascade)	891.97	893	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
B19	II	m_t/m_c	not documented (legacy label: cascade)	135.49	136	Legacy value; derivation not documented in the current dataset; retained for traceability, not counted as a prediction
C01	III	m_τ/m_e (dyn)	Wilson line 3-gen optimization	3403.1	3477.37	conditional reconstruction (calibration choices, main §6.2)
C02	III	m_τ/m_μ (dyn)	Wilson line 3-gen optimization	16.5441	16.8177	conditional reconstruction (calibration choices, main §6.2)
C03	III	c*	Non-adiabatic threshold	0.177828	n/a	diagnostic of the approximate local metric
C04	III	$\sin^2\theta_W$ (RGE)	2-loop MSSM RGE	0.237655	0.23122	retired (supersymmetric running from a GUT boundary; main App. D)
C05	III	α_s (RGE)	2-loop MSSM RGE	0.103769	0.118	retired (supersymmetric running from a GUT boundary; main App. D)
C06	III	M_{GUT}	Gauge unification	2e+16	2e+16	retired (supersymmetric running from a GUT boundary; main App. D)
C07	III	${\alpha_{GUT}}^{-1}$	Gauge unification	25.3	25.3	retired (supersymmetric running from a GUT boundary; main App. D)
C08	III	λ_1	Neumann FD + Richardson	0.124607	n/a	diagnostic of the approximate local metric
C09	III	α_{Weyl}	$N(\lambda) \sim C\ \lambda^{d/2}$	3.46002	3.5	diagnostic of the approximate local metric (compared with d/2, not a measurement)
C10	III	N_{KK}	Unified KK tower enumeration	22671	n/a	diagnostic of the approximate local metric
C11	III	N_{fiber}	Adiabatic + explicit fiber modes	57578	n/a	diagnostic of the approximate local metric
C12	III	spacing	Level spacing statistics	1.35838	n/a	diagnostic of the approximate local metric
C13	III	κ_{gauge}	f_{IJ} eigenvalue spread	1.0471	1	diagnostic of the approximate local metric (compared with 1, not a measurement)
C14	III	n_{SD}	Count of positive eigenvalues of the K3 intersection form	3	3	diagnostic of the approximate local metric (compared with N_{gen})
C15	III	N_{assoc}	Homology 3-cycles with φ-calibration	57	n/a	diagnostic of the approximate local metric
C16	III	$\|\|T\|\|_C{}^0$	G_2 torsion norm	2.98e-05	n/a	diagnostic of the approximate local metric
C17	III	ΔV(e-τ)	Associative volume differences	8.6331	8.15393	conditional reconstruction (calibration choices, main §6.2)
C18	III	ΔV(e-μ)	Associative volume differences	3.27125	2.82257	conditional reconstruction (calibration choices, main §6.2); mismatched reference (see Type III above)
C19	III	m_τ/m_e (comb)	Wilson-line ratio × $\exp(e^K \cdot \Delta V)$	3485.3	3477.37	conditional reconstruction (calibration choices, main §6.2)

ID	Type	Quantity	Formula or mechanism	Framework value	Reference value	Status
C20	III	m_τ/m_μ (comb)	Wilson-line ratio × $\exp(e^K \cdot \Delta V)$	16.69	16.8177	conditional reconstruction (calibration choices, main §6.2); inherits the C18 pairing
C21	III	m_μ/m_e (comb)	Wilson-line ratio × $\exp(e^K \cdot \Delta V)$	208.8	206.768	conditional reconstruction (calibration choices, main §6.2); ratio C19/C20, inherits the C18 pairing
D01	IV	b_2	TCS K_7	21	n/a	input (R1)
D02	IV	b_3	TCS K_7	77	n/a	input (R1)
D03	IV	H*	b_2+b_3+1	99	n/a	integer combination of R1
D04	IV	N_{gen}	$b_2/\dim(K_7)$	3	n/a	integer identity (main §2.4)
D05	IV	det(g)	$\det(g_5)$	2.03125	n/a	imposed normalization (R2)
D06	IV	h_{NK}	$\beta\times\eta\times\gamma$	8.95e-09	n/a	diagnostic of the approximate local metric
D07	IV	dist(g,g*)	NK contraction ball	4.86e-06	n/a	diagnostic of the approximate local metric
D08	IV	$\|\|T\|\|_C{}^0$ (cert)	Chebyshev coefficient bound	2.95e-05	n/a	diagnostic of the approximate local metric
D09	IV	$\kappa(G_{K3})$	L^2 Gram matrix K3 2-forms	1.05232	n/a	diagnostic of the approximate local metric
D10	IV	N_{levels}	Unified spectrum enumeration	22671	n/a	diagnostic of the approximate local metric
D11	IV	`N_states_total`	Including degeneracies	210060	n/a	diagnostic of the approximate local metric
D12	IV	V_{eff}	Effective 7D volume from Weyl fit	5.38e+05	n/a	diagnostic of the approximate local metric
D13	IV	$n_{homology}$	$b_3 = 77$	77	n/a	diagnostic of the approximate local metric
D14	IV	N_{assoc}	φ-calibrated cycles	57	n/a	diagnostic of the approximate local metric
D15	IV	$\|\|Y\|\|$	Total Yukawa coupling norm	8.96902	n/a	diagnostic of the approximate local metric
D16	IV	ΔV_{range}	max(V) - min(V)	8.91952	n/a	diagnostic of the approximate local metric
D17	IV	g_{ss}	$g_{ss} = 19/6$ (S^1 sector)	3.16667	n/a	diagnostic of the approximate local metric
D18	IV	g_{T2}	$g_{T2} = 7/6$ (T^2 sector)	1.16667	n/a	diagnostic of the approximate local metric
D19	IV	g_{K3}	$g_{K3} \sim 64/77$ (K3 sector mean)	0.831169	n/a	diagnostic of the approximate local metric
D20	IV	N_{QNM}	b2 + b3 = H* - 1	98	n/a	from [42]; topological count
D21	IV	`b3/b3_S3S4`	b3(K7) = 77 vs b3(S3xS4) = 1	77	n/a	from [42]; topological count
D22	IV	M_{res}	$\tau_0 = v_{EW}$, $M_{res} = v_{EW}{}^2 / M_{Pl}$	4969.2	n/a	from [42]; conditional on the scale identification $\tau_0 = v_{EW}$

No entry is ranked by its deviation. An agreement between a Type II number and a PDG mass quoted to two or three figures measures the rounding, and a mean deviation measures closeness, which main §7 shows is not distinctive.

S3.7 Lean cross-references

The Lean formalization (Arithmon/K7-Lean, main §8) verifies arithmetic identities. A verified identity says that a closed form evaluates to a number; it says nothing about the physical identification, the choice of formula, or the consistency of relations taken together. The table gives, for each of the 33 relations, the Lean declaration that ties the closed form to its value, and the scope of that proof. It is generated from `canonical/results/k7_v3_6_lean_map.json`; the library builds at the commit named, and the axioms are those reported by `#print axioms`.

Arithmon/K7-Lean, commit `34ce857` (merged into `main`, PR #167); aggregate `Certificate.Relations V36.relations_v36_certified`, which depends only on the axioms `propext`, `Classical.choice`, `Quot. sound`.

ID	Observable	Lean declaration (under `GIFT.`)	Location	Scope
A01	$\sin^2\theta_{\mathrm{W}}$	`Certificate.RelationsV36.A01_sin2_theta_W`	Certificate/ RelationsV36.lean:50	exact rational equality of the closed form
A02	$\alpha_{\mathrm{s}}(\mathrm{M_Z})$	`Certificate.RelationsV36.A02_alpha_s`	Certificate/ RelationsV36.lean:55	closed form in $\mathbb{R}$ equal to the stated real value (no numerical bound)
A03	λ_{H}	`Certificate.RelationsV36.A03_lambda_H`	Certificate/ RelationsV36.lean:59	closed form in $\mathbb{R}$ equal to the stated real value (no numerical bound)
A04	$\alpha^{-1}(0)$	`Certificate.RelationsV36.A04_alpha_inv`	Certificate/ RelationsV36.lean:65	exact rational equality of the closed form
A05	$\mathrm{Q_{Koide}}$	`Certificate.RelationsV36.A05_Q_Koide`	Certificate/ RelationsV36.lean:73	exact rational equality of the closed form
A06	$\mathrm{m_\tau/m_e}$	`Certificate.RelationsV36.A06_m_tau_m_e`	Certificate/ RelationsV36.lean:77	exact rational equality of the closed form
A07	$\mathrm{m_\mu/m_e}$	`Certificate.RelationsV36.A07_m_mu_m_e`	Certificate/ RelationsV36.lean:81	base 27 and interval bounds on 27^{φ}
A08	$\mathrm{m_s/m_d}$	`Certificate.RelationsV36.A08_m_s_m_d`	Certificate/ RelationsV36.lean:91	exact rational equality of the closed form
A09	$\mathrm{m_c/m_s}$	`Certificate.RelationsV36.A09_m_c_m_s`	Certificate/ RelationsV36.lean:95	exact rational equality of the closed form
A10	$\mathrm{m_b/m_t}$	`Certificate.RelationsV36.A10_m_b_m_t`	Certificate/ RelationsV36.lean:100	exact rational equality of the closed form
A11	$\mathrm{m_u/m_d}$	`Certificate.RelationsV36.A11_m_u_m_d`	Certificate/ RelationsV36.lean:104	exact rational equality of the closed form
A12	$\sin^2\theta_{12}{}^{\mathrm{PMNS}}$	`Certificate.RelationsV36.A12_sin2_theta12_PMNS`	Certificate/ RelationsV36.lean:109	exact rational equality of the closed form
A13	$\sin^2\theta_{23}{}^{\mathrm{PMNS}}$	`Certificate.RelationsV36.A13_sin2_theta23_PMNS`	Certificate/ RelationsV36.lean:113	exact rational equality of the closed form
A14	$\sin^2\theta_{13}{}^{\mathrm{PMNS}}$	`Certificate.RelationsV36.A14_sin2_theta13_PMNS`	Certificate/ RelationsV36.lean:117	exact rational equality of the closed form
A15	δ_{CP}	`Certificate.RelationsV36.A15_delta_CP`	Certificate/ RelationsV36.lean:122	the integer 197 (the unit is not in Lean)
A16	$\sin\theta_{12}{}^{\mathrm{CKM}}$	`Certificate.RelationsV36.A16_sin_theta12_CKM`	Certificate/ RelationsV36.lean:126	exact rational equality of the closed form
A17	$\mathrm{A_{Wolf}}$	`Certificate.RelationsV36.A17_A_Wolf`	Certificate/ RelationsV36.lean:131	exact rational equality of the closed form

ID	Observable	Lean declaration (under `GIFT.`)	Location	Scope
A18	$\sin\theta_{23}{}^{\mathrm{CKM}}$	`Certificate.RelationsV36.A18_sin_theta23_CKM`	Certificate/RelationsV36.lean:135	exact rational equality of the closed form
A19	m_H/m_t	`Certificate.RelationsV36.A19_m_H_m_t`	Certificate/RelationsV36.lean:140	exact rational equality of the closed form
A20	m_H/m_W	`Certificate.RelationsV36.A20_m_H_m_W`	Certificate/RelationsV36.lean:145	exact rational equality of the closed form
A21	m_W/m_Z	`Certificate.RelationsV36.A21_m_W_m_Z`	Certificate/RelationsV36.lean:150	exact rational equality of the closed form
A22	m_μ/m_τ	`Certificate.RelationsV36.A22_m_mu_m_tau`	Certificate/RelationsV36.lean:155	exact rational equality of the closed form
A23	Ω_{DE}	`Certificate.RelationsV36.A23_Omega_DE`	Certificate/RelationsV36.lean:159	closed form in $\mathbb{R}$ equal to the stated real value (no numerical bound)
A24	n_s	`Certificate.RelationsV36.A24_n_s`	Certificate/RelationsV36.lean:165	indices (11, 5) and $\zeta(D_{\mathrm{bulk}})/\zeta(\mathrm{Weyl}) = \zeta(11)/\zeta(5)$ (no numerical bound)
A25	$\Omega_{\mathrm{DM}}/\Omega_b$	`Certificate.RelationsV36.A25_Omega_DM_b`	Certificate/RelationsV36.lean:173	exact rational equality of the closed form
A26	Ω_c/Ω_Λ	`Certificate.RelationsV36.A26_Omega_c_Lambda`	Certificate/RelationsV36.lean:177	exact rational equality of the closed form
A27	Ω_Λ/Ω_m	`Certificate.RelationsV36.A27_Omega_Lambda_m`	Certificate/RelationsV36.lean:182	exact rational equality of the closed form
A28	h	`Certificate.RelationsV36.A28_hubble_h`	Certificate/RelationsV36.lean:187	exact rational equality of the closed form
A29	Ω_b/Ω_m	`Certificate.RelationsV36.A29_Omega_b_m`	Certificate/RelationsV36.lean:191	exact rational equality of the closed form
A30	Ω_c/Ω_m	`Certificate.RelationsV36.A30_Omega_c_m`	Certificate/RelationsV36.lean:196	exact rational equality of the closed form
A31	σ_8	`Certificate.RelationsV36.A31_sigma_8`	Certificate/RelationsV36.lean:200	exact rational equality of the closed form
A32	Y_p	`Certificate.RelationsV36.A32_Y_p`	Certificate/RelationsV36.lean:204	exact rational equality of the closed form
A33	Ω_m/Ω_Λ	`Certificate.RelationsV36.A33_Omega_m_Lambda`	Certificate/RelationsV36.lean:208	exact rational equality of the closed form

Scope. Every relation has a statement that ties its closed form, written in the structural constants, to its value: an exact rational equality for 27; a closed form in $\mathbb{R}$ equal to the stated real value for α_s, λ_H and Ω_{DE}, with no numerical bound; the indices and the ζ-quotient for n_s, with no numerical bound; the base 27 and the bounds $206 < 27^{\varphi} < 209$ for m_μ/m_e, an interval wider than the deviation it is compared with; and the integer 197 for δ_{CP}, whose unit is a dictionary assignment outside Lean. The 33 statements are collected in `relations_v36_certified`.

Names. Declarations are named for what they state: the CKM entries are `sin_theta12_CKM` and `sin_theta23_CKM` (the quantities compared are sin θ, S2.4); 42 is `two_b2 := 2 * b2` ($\chi(K_7) = 0$, S1 §9.3); identities giving other values for the paper's observables (4/7 and 8/14 for θ_{23}, 7/23 for θ_{12}, 85/99) carry neutral arithmetic names and are instances of the vocabulary freedom of S2.7. Earlier names are kept as deprecated aliases. The factor 62/69 once applied to δ_{CP} ($197 \times 62/69 = 12214/69$, S2.8) is not in the

aggregated certificate; its module is kept, marked withdrawn. A statement proves that the Koide parameter built from the framework's own $m_\mu/m_e = 27^\varphi$ and $m_\tau/m_e = 3477$ lies strictly below 2/3 (`Relations/Koide Assembly.lean`), the incompatibility of main §6.1.

Axioms. The formalization rests on fourteen axioms beyond Mathlib: four that import results from the literature (`cheeger_inequality`, `literature_package`, `spectral_upper_bound`, `neck_dominates`) and ten interval certificates that import numerical enclosures [ERR, §7.1]; no `sorry` is used.

Changes since version 3.5

The sections above state the present position. This table lists, once, what earlier versions of this supplement and of the dataset said.

Earlier statement	Present position	Section
the dataset is "95 observables derived from a single G_2 metric, zero freely adjustable parameters"	33 relations assigned by the author, in which the metric does not enter; 62 entries not counted	S3.1
the nineteen Type II entries are each "a Type I ratio times an experimental anchor"	unit assignments, scale bridges and legacy values	S3.2
four entries from a two-loop supersymmetric running from a grand-unification boundary	retired with the running	S3.2
density ratios propagated without correlations: $\Omega_c/\Omega_m = 0.839(11)$, $+0.4\sigma$	computed on the Planck chains: 0.8390(19), $+2.5\sigma$	S3.3
m_W/m_Z compared with the ratio line of the PDG summary table, 0.88136(15), -2.7σ	computed from the published masses like every other mass ratio: 0.88128(9), -3.8σ	S3.3

Earlier statement	Present position	Section
reference data of the earlier dataset: uncertainties of m_μ/m_τ and m_τ/m_e larger than the PDG errors propagate (by factors 13 and 1.3); Q_{Koide} computed from the PDG 2022 τ mass while m_τ/m_e used the later one; an uncertainty of m_b/m_t with no source; an h that matched no Planck column; NuFIT variants with and without atmospheric data mixed; an uncertainty of 0.02 for m_μ/m_e where CODATA 2022 gives 4.6×10^{-6}; the two CKM entries labelled $\sin^2\theta$ although their values are $\sin\theta_{12} = \lambda$ and $\sin\theta_{23} = A\lambda^2$. On those uncertainties 28 relations lay within 2σ	corrected data and one convention for all uncertainties; classes 23 / 2 / 6 / 1 / 1	S3.3
Y_p labelled Planck 2018	the PDG's observational helium abundance	S3.3
$N_{gen} = 3$ as the output of an index theorem; b_2, b_3 confirmed by the spectral analysis of [B]	integer identity; posited inputs	S3.4
"exact matches" (deviation below 0.01%) and a ranking of sectors by mean deviation; M_{GUT}, ${\alpha_{GUT}}^{-1}$ and rank(Y) compared with values the same computation had set; Q_{Koide} as "the most precise agreement in the framework, with no fitting involved"; a table of neutrino angles with formulas ($\theta_{23} = \arcsin(75/99)$, θ_{12} from an auxiliary parameter) that are not the dataset's	no ranking by deviation; $Q = 2/3$ predates the framework and every invented vocabulary reaches it (main §6.1); the dataset's relations are those of main §4.1	S3.6
the framework values of m_u and m_c stored rounded (2.16, 1280); a summary count of "exact" entries	the values of their formulas (2.160247, 1280.26); the count is removed	S3.6

Earlier statement	Present position	Section
the single theorem `observables_certified` (`Certificate/Predictions.lean`) cited as certifying the relations; it is a conjunction of statements `X = v` in which `X` is defined as the literal `v`. Before the alignment of 2026-10-22 (commit `70dc589`), 24 relations had an exact statement elsewhere in the library, 5 had only their components, 1 had bounds and 3 (m_μ/m_τ, Ω_c/Ω_m, Ω_m/Ω_Λ) had none	every relation has a statement tying its closed form to its value, collected in `relations_v36_certified`	S3.7

S3.8 Data availability

The dataset is `canonical/data/observables.json` (95 entries; for each: formula, framework value, reference value and uncertainty, source, deviation, Lean theorem, provenance, and dated notes of every change). The refresh script and its log, and the generators of the main §4.1 table and of the S3 tables, are listed in S3.3 and S3.5. The dataset is deposited in the public repository (Arithmon/K7), as read by the look-elsewhere run in `publications/validation/v3.6/canonical/data/observables.json` and as a table in `publications/references/observables.csv`. The refresh script and the table generators are available from the author on request. The spectral companion artifacts of [B] are archived on Zenodo (DOI: 10.5281/zenodo.18920367).

Author's Related Works

- **[B]** B. de La Fournière, "Spectral Geometry of an Explicit G_2 Metric: Laplacian Spectrum and Harmonic Forms," Zenodo 10.5281/zenodo.18920367 (2026).
- **[ERR]** B. de La Fournière, "Erratum: GIFT / K_7 framework v3.5" (2026), repository Arithmon/K7, `publications/ERRATUM_v3.5.md`, commit aed460315eecc17a20402b1f4d20ac310dd4f6c2, github.com/Arithmon/K7/blob/aed4603/publications/ERRATUM_v3.5.md.

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